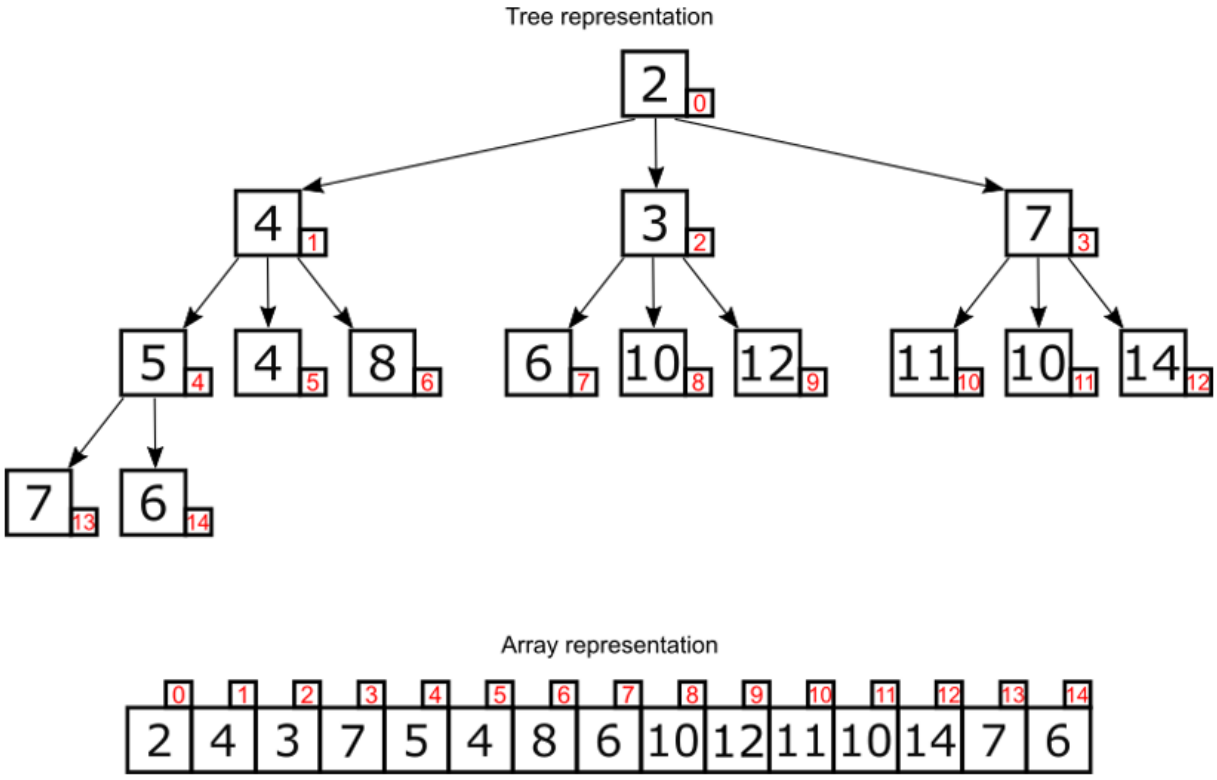


Figure 2.5 A 3-way heap. Only priorities are shown for the nodes (values are irrelevant here); in red, the indices of the heap elements in the array).
Top: the tree representation for the heap. Notice how every parent is smaller (or at most equal) of its children and, in turn, all elements in its subtree.
Bottom: the same heap in its array representation.

Figure 2.5 shows both the tree and array representation of a 3-ary heap. The same idea holds for branching factor 4, 5 and so on.

For a *d-ary* heap, where the branching factor is the integer `D > 1`, our three heap invariants become:

1. Every node has at most `D` children.